

TRIANGLE NORTH EXECUTIVE, NC, USA

WMO: 722141

Lat: 36.023N

Lon: 78.330W

Elev: 113

StdP: 99.98

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03731

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -5.0	(d) -17.6	(e) 0.8	(f) -4.2	(g) -15.1	(h) 1.0	(i) -1.8	(j) 7.7	(k) 14.3	(l) 6.9	(m) 11.7	(n) 1.1	(o) 310	(p) 0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.4	24.0	32.9	23.7	32.2	23.4	25.8	31.2	25.2	30.6	24.6	29.7	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.8	28.2	23.7	18.8	27.5	22.9	17.9	26.6	80.1	31.5	77.6	30.1	75.0	30.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.4	4.7	-12.2	36.5	2.9	1.9	-14.2	37.8	-15.9	39.0	-17.5	40.0	-19.6	41.4
			-13.0	27.2	2.6	1.0	-14.9	28.0	-16.4	28.6	-17.8	29.1	-19.7	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.9	6.7	10.2	15.6	19.9	24.1	25.9	25.3	22.1	16.0	10.1	7.0
	DBStd	8.65	5.76	5.54	5.50	4.78	3.92	3.05	2.39	2.66	3.08	4.55	4.90	5.25
	HDD10.0	567	177	121	69	9	0	0	0	0	0	6	59	126
	HDD18.3	1853	416	326	258	109	32	2	0	0	7	101	250	354
	CDD10.0	2644	19	29	74	176	306	424	493	474	363	192	61	32
	CDD18.3	889	0	1	4	26	79	176	235	216	119	29	2	2
	CDH23.3	7933	0	7	48	248	658	1658	2274	1944	872	208	11	4
	CDH26.7	3104	0	0	5	54	194	681	1009	814	298	48	0	0
Wind	WSAvg	1.6	1.8	2.0	2.1	2.2	1.6	1.3	1.3	1.2	1.4	1.4	1.6	1.6
Precipitation	PrecAvg	1153	96	89	103	79	102	94	116	130	103	84	80	79
	PrecMax	1654	199	198	208	210	261	253	260	309	529	291	199	186
	PrecMin	869	24	19	25	9	39	11	31	12	16	0	17	9
	PrecStd	198	42	44	42	42	48	59	52	71	85	56	49	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	24.2	27.4	30.5	32.6	36.2	36.8	36.2	33.7	31.0	24.9	22.9
		MCWB	15.8	16.8	16.6	18.0	22.1	23.1	24.5	25.2	22.3	22.3	17.3	18.4
	2%	DB	18.9	21.4	24.7	28.0	30.9	33.7	34.5	33.8	32.0	27.7	22.7	21.1
		MCWB	15.0	15.1	15.6	18.0	21.5	23.3	24.5	24.3	22.5	20.8	16.6	18.1
	5%	DB	17.2	18.9	22.4	26.3	28.8	32.3	32.9	32.5	30.1	25.7	20.9	18.2
		MCWB	13.5	13.6	14.7	17.3	21.2	23.1	24.2	23.9	22.2	20.0	16.6	15.9
	10%	DB	14.8	16.9	20.0	24.0	27.3	30.8	32.1	31.2	28.0	23.4	18.0	16.2
		MCWB	11.4	12.2	14.1	16.9	20.1	22.8	23.9	23.6	21.6	19.1	14.7	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.7	19.0	22.0	24.0	25.8	27.1	27.0	25.5	24.3	20.3	20.3
		MCDB	18.9	22.1	23.6	25.1	28.7	30.5	32.7	32.8	28.4	27.4	21.6	22.0
	2%	WB	16.1	17.0	17.7	20.3	22.9	24.9	25.9	25.8	24.5	22.9	18.8	18.4
		MCDB	18.3	19.4	21.1	24.4	27.6	30.7	31.9	31.1	28.3	25.5	21.1	20.2
	5%	WB	14.0	15.2	16.7	18.9	22.2	24.2	25.2	25.2	23.7	21.9	17.2	16.7
		MCDB	16.5	18.8	20.3	23.8	26.9	30.0	31.2	30.2	27.4	24.6	19.3	18.2
	10%	WB	11.6	12.4	15.2	17.9	21.3	23.5	24.7	24.5	22.9	20.2	15.5	13.8
		MCDB	13.8	15.6	19.1	22.1	25.5	28.7	30.0	29.2	26.2	22.7	18.3	16.0
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	12.4	12.9	11.6	11.3	10.9	10.5	10.5	11.8	11.9	10.6
		MCDBR	13.2	13.8	14.5	14.8	12.9	12.5	11.9	11.9	12.7	13.0	13.6	11.9
		MCWBR	9.9	9.0	7.8	6.6	5.4	4.6	4.2	4.1	5.1	7.0	8.6	9.1
	5% WB	MCDBR	12.0	13.0	12.3	11.6	10.8	11.1	10.6	10.3	10.0	10.3	10.9	10.8
		MCWBR	10.4	9.9	8.2	6.2	5.2	4.7	4.3	4.2	4.8	6.2	8.7	9.3
Clear-Sky Solar Irradiance	taub	0.310	0.327	0.357	0.403	0.452	0.498	0.534	0.528	0.416	0.369	0.328	0.312	
	taud	2.504	2.451	2.404	2.275	2.172	2.088	2.005	2.007	2.319	2.431	2.504	2.535	
	Ebn at Noon	890	913	911	880	839	797	765	760	839	852	861	865	
	Edh at Noon	81	96	110	132	149	162	175	171	119	98	81	75	
All-Sky Solar Radiation	RadAvg	2.54	3.23	4.25	5.38	5.96	6.41	6.06	5.47	4.53	3.66	2.79	2.21	
	RadStd	0.18	0.37	0.31	0.40	0.46	0.43	0.34	0.29	0.38	0.46	0.23	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+106

Nomenclature: See separate page